

Vincent Schüßler

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Experience

- Wētā FX**, Visiting Researcher Jan 2026 – July 2026
- Developing sampling techniques for volumetric path tracing
 - Implementation in the Manuka production renderer (C++)
- KIT Computer Graphics Group**, Researcher Feb 2019 – Jan 2025
- Conducted and published research projects
 - Developed and contributed to research rendering software (Rust, C)
 - Advised students for theses and courses

Education

- Dr.-Ing. Computer Graphics Group**, Karlsruhe Institute of Technology (KIT) Feb 2019 – July 2026
(PhD) My research focused on the efficient and high-quality rendering of complex light transport phenomena. Other interests include real-time GPU path tracing, texturing, filtering and microfacet models.
- Thesis: *Guided Sampling and Caching for Efficient Physically-Based Rendering*
 - Expected completion in July 2026 (defense scheduled)
- M.Sc. Computer Science**, Karlsruhe Institute of Technology (KIT) Apr 2016 – Nov 2018
- Thesis: *High-Dimensional Path Guiding* (computer graphics)
 - Areas of specialization: Computer graphics and machine learning
- B.Sc. Computer science**, Karlsruhe Institute of Technology (KIT) Oct 2012 – Feb 2016
- Thesis: *Geometrically Correct Normal Maps Using Non-Symmetric Microsurfaces* (computer graphics)

Projects

- Demoscene activity as member of rtifical** 2013 – 2017
- Developed shaders for SDF-based rendering and post processing (GLSL)
 - Contributed to custom tools for visual sequencing and real-time music synthesis (C++)

Skills

Programming languages: Rust, Python, C, C++, GLSL, JavaScript
Technologies: Vulkan, Git, Linux, JAX
Languages: English (fluent), German (native)

Publications

- Selective Caching in Procedural Texture Graphs for Path Tracing** July 2025
Vincent Schüßler, Johannes Hanika, Basile Sauvage, Jean-Michel Dischler, Carsten Dachsbacher
Eurographics Symposium on Rendering
<https://doi.org/10.2312/sr.20251176>
- Sampling of Anisotropic Spatial Gaussians for Path Guiding** May 2025
Sergey Lelyakin, Vincent Schüßler, Carsten Dachsbacher
Eurographics 2025 - Posters
[10.2312/egp.20251017](https://doi.org/10.2312/egp.20251017)

ReSTIR Subsurface Scattering for Real-Time Path Tracing

Aug 2024

Mirco Werner, Vincent Schüßler, Carsten Dachsbacher

Proc. ACM CGIT (Proc. High Performance Graphics)

[10.1145/3675372](#)**Bridge Sampling for Connections via Multiple Scattering Events**

July 2024

Vincent Schüßler, Johannes Hanika, Carsten Dachsbacher

Computer Graphics Forum (Proc. Eurographics Symposium on Rendering)

[10.1111/cgf.15160](#)**Markov Chain Mixture Models for Real-Time Direct Illumination**

July 2023

Addis Dittebrandt, Vincent Schüßler, Johannes Hanika, Sebastian Herholz, Carsten Dachsbacher

Computer Graphics Forum (Proc. Eurographics Symposium on Rendering)

[10.1111/cgf.14881](#)**Path Guiding with Vertex Triplet Distributions**

July 2022

Vincent Schüßler, Johannes Hanika, Alisa Jung, Carsten Dachsbacher

Computer Graphics Forum (Proc. Eurographics Symposium on Rendering)

[10.1111/cgf.14582](#)**Microfacet-based normal mapping for robust Monte Carlo path tracing**

Nov 2017

Vincent Schüßler, Eric Heitz, Johannes Hanika, Carsten Dachsbacher

Transactions on Graphics (Proc. SIGGRAPH Asia)

[10.1145/3130800.3130806](#)